AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 24 · STUDENT LESSON

Feedback Loops

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 19: collected observations depend on who or what was selected.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Implement a deterministic recommender. Try the recall before opening notes.

Your learning target

I can implement a **deterministic recommender**.

I can handle ties and no eligible options.

I can explain recommendations with their limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your **explanation** that the **evidence** shows was wrong.

Week 24: **Feedback Loop**s

Recommendations influence behavior, which then becomes new data.



THINKER MISSION

Map how repeated clicks can narrow future suggestions.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> An option violates a hard duration limit but has the best score. Should it win?

This week's end goal

By the end, I can explain how recommendation **exposure** can create a **feedback loop**.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Feedback Loops case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **feedback loop**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields.

STUDENT LESSON · 3 OF 18

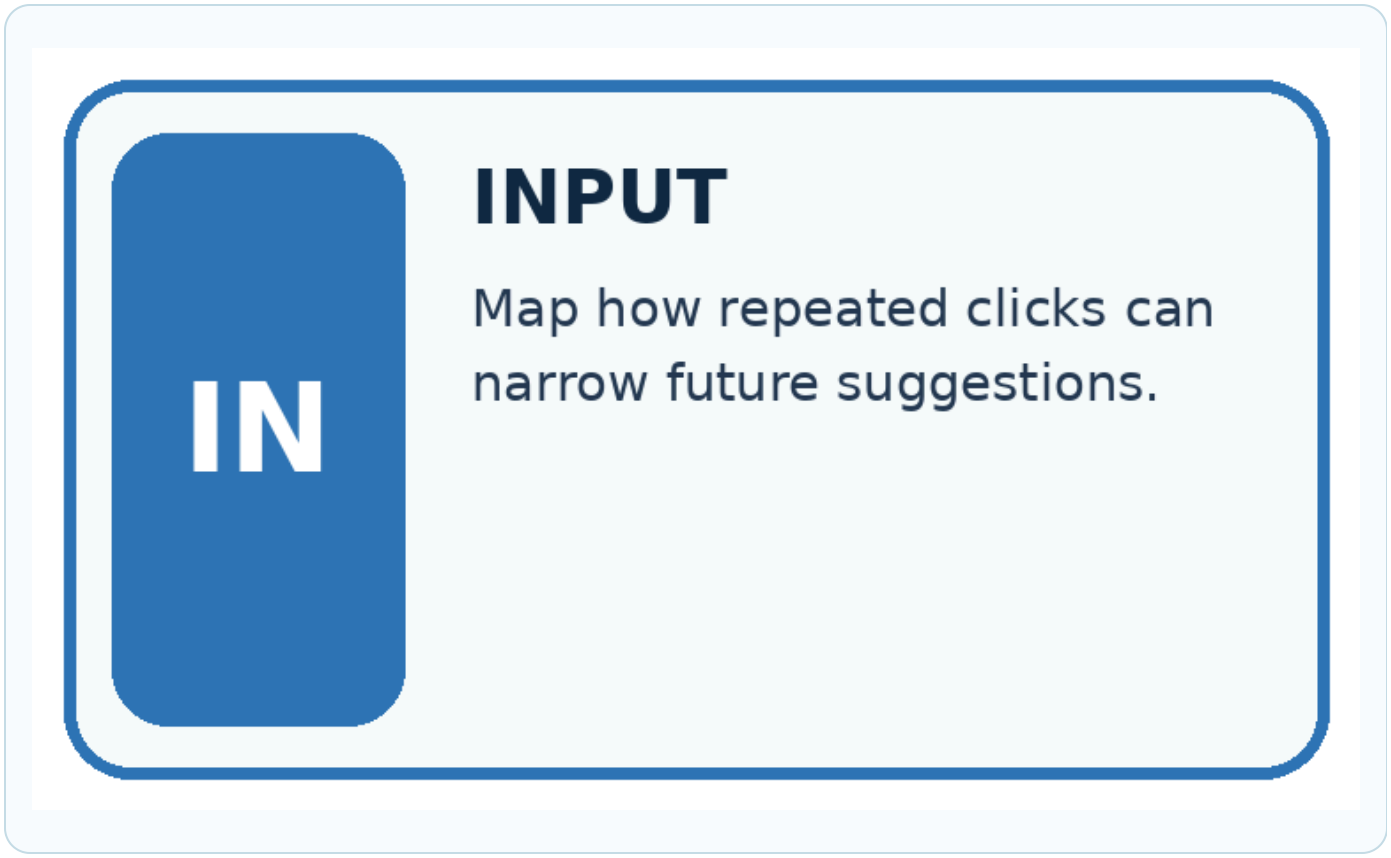
See how it works

> Follow the mechanism for Feedback Loops. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

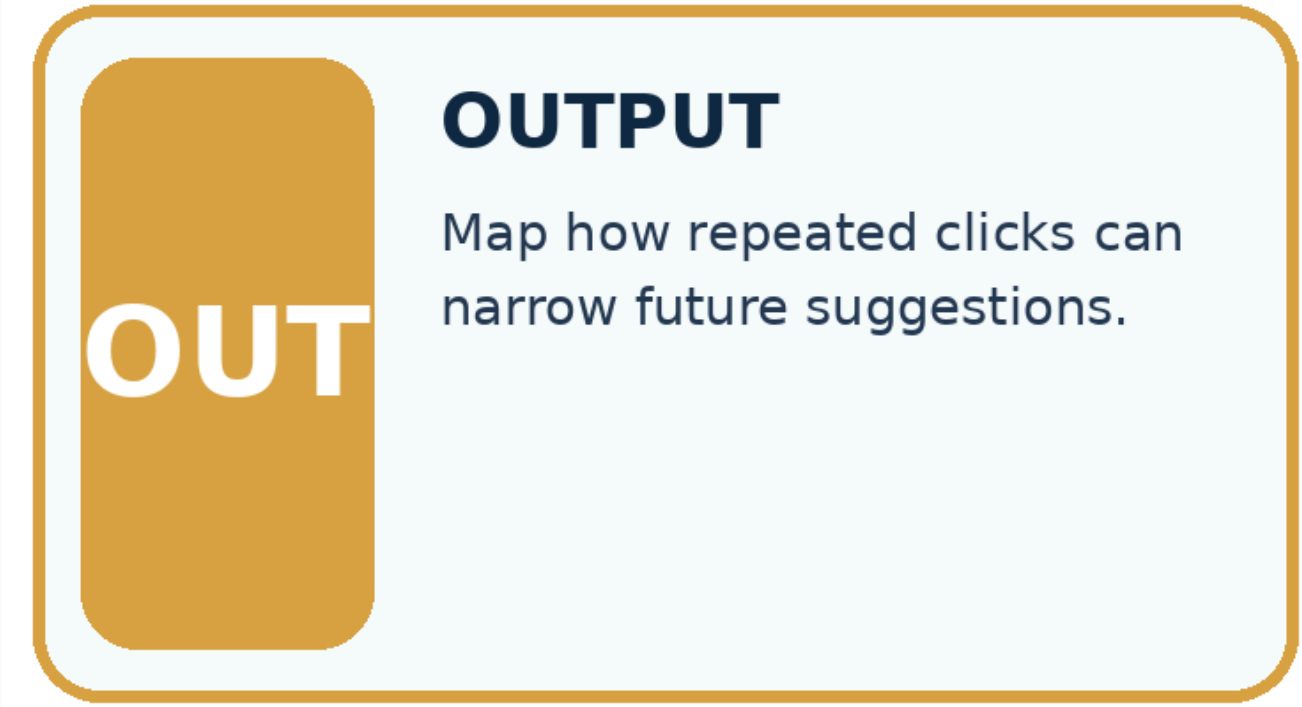
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Map how repeated clicks can narrow future suggestions.

↓


PROCESS
Recommendations influence behavior, which then becomes new data.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
def recommend(items, limit):
    best = None
    best_score = None
    for item in items:
        if item["minutes"] <= limit:
            score = 2 * item["fun"] + item["learning"]
            if (best is None or score > best_score
                or (score == best_score and item["name"] < best)):
```

```
                best = item["name"]
                best_score = score
    return best

items = [{"name": "B", "fun": 3, "learning": 2, "minutes": 10},
         {"name": "A", "fun": 2, "learning": 4, "minutes": 9}]
print(recommend(items, 10))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
B	Eligible score 8	Temporary best B
A	Eligible score 8	Alphabetical tie gives A

Read each row from left to right and connect the evidence to the stated rule.

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run.

After selecting an item, best_score is numeric. An empty list or no eligible item returns None.

Trace the best-so-far record

Write a trace table with item, eligible, score and best-so-far. In the worked list, B qualifies and scores 8, so B becomes the current best. A also qualifies and scores 8.

The scores tie, so the written alphabetical rule changes the best to A. Starting with A gives the same final answer. If no item qualifies, best stays None; do not invent a winner or quietly relax the limit.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

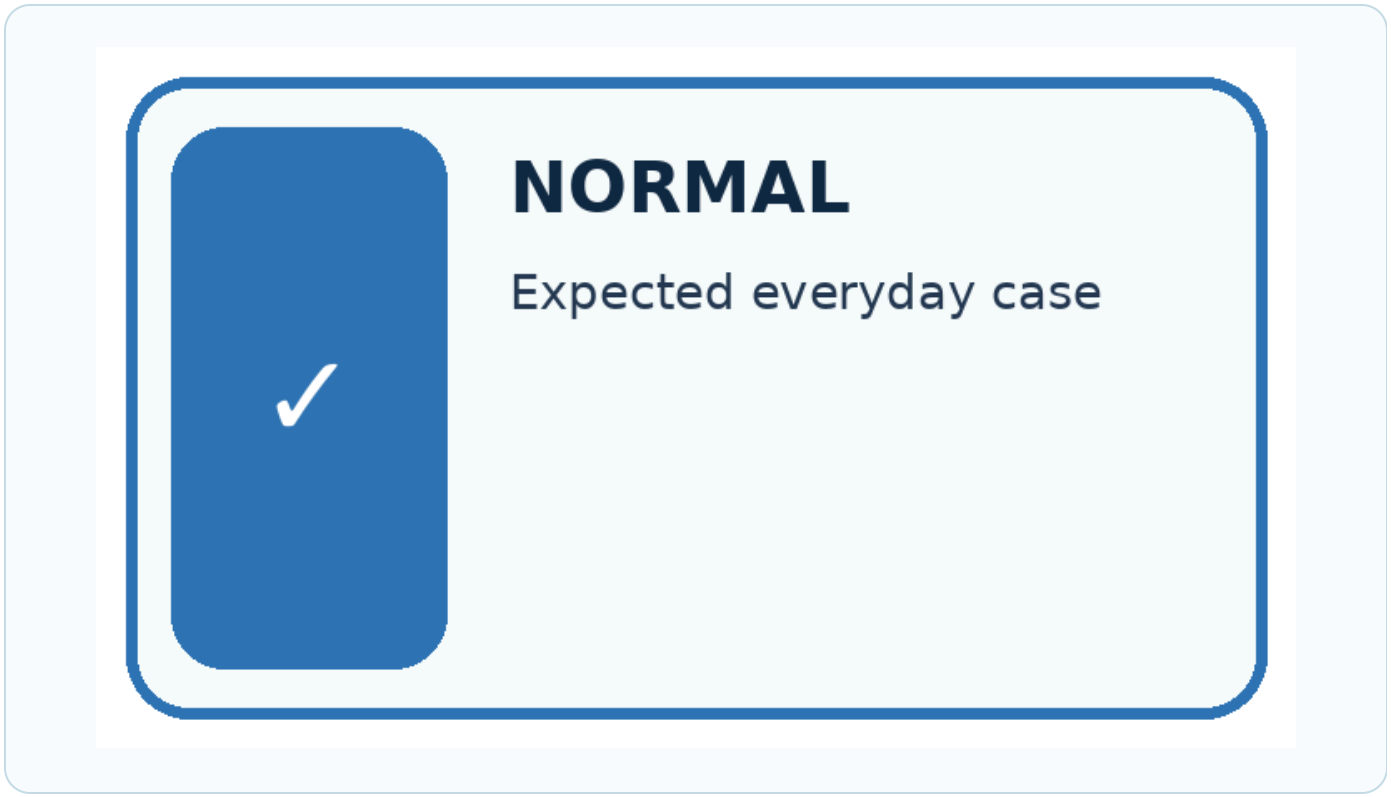
> Compare the supplied cases for Feedback Loops. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

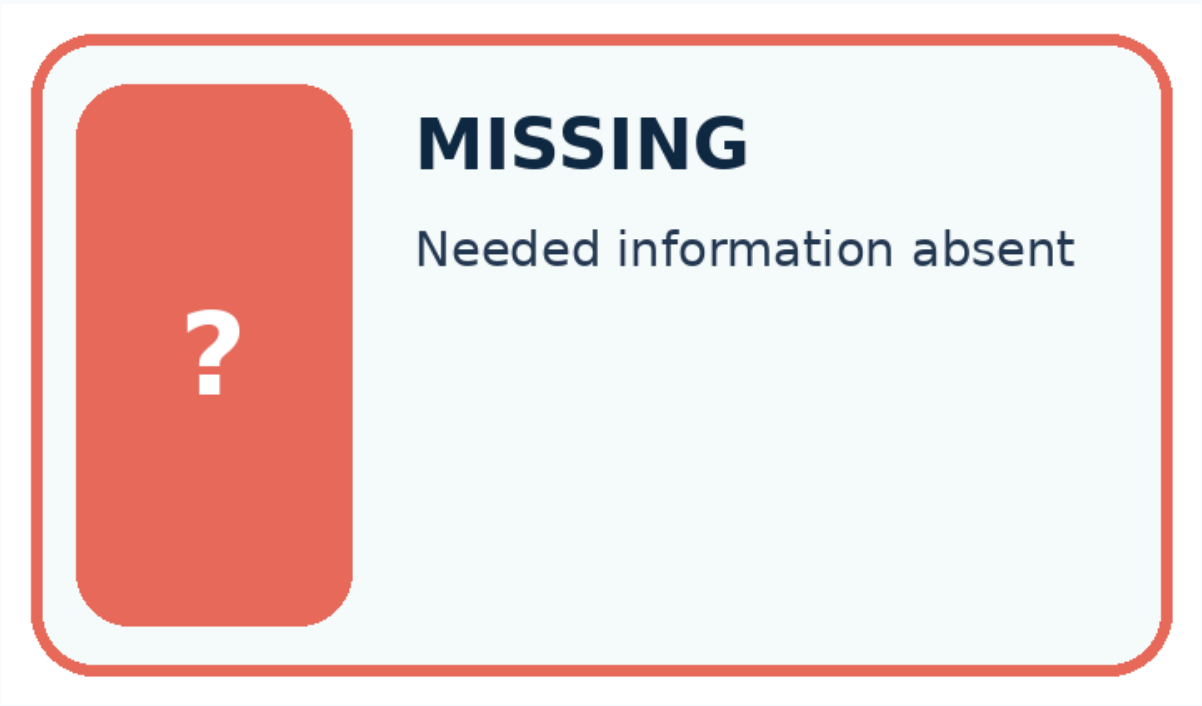
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: On a planned trial, show B once and record no click.

Predicted result

Prediction: B gains one **exposure** but still zero clicks.

Why the result follows

Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

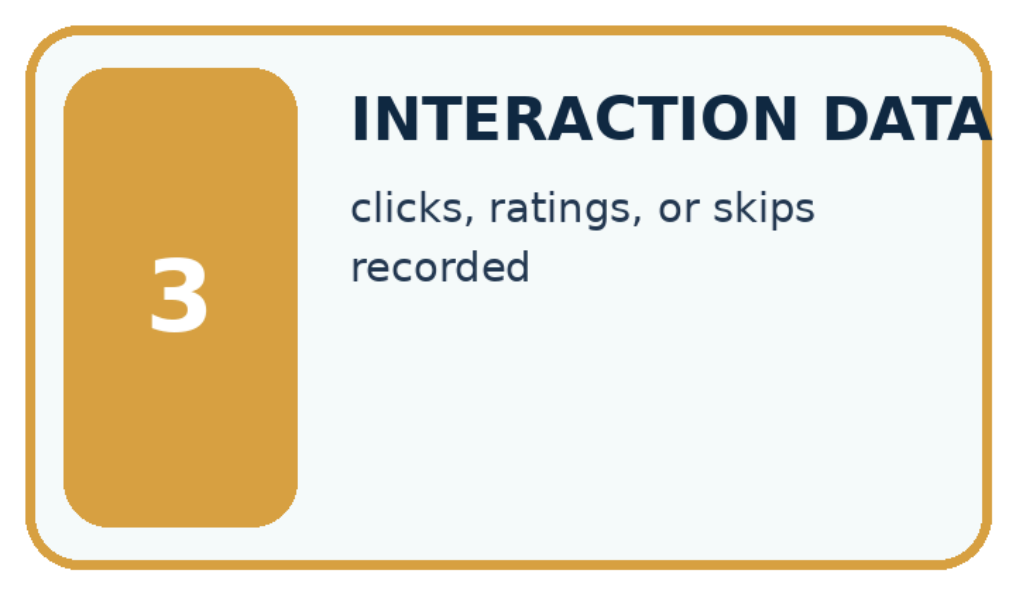
> Use the vocabulary of Feedback Loops to explain a concrete decision, rather than only memorising definitions.

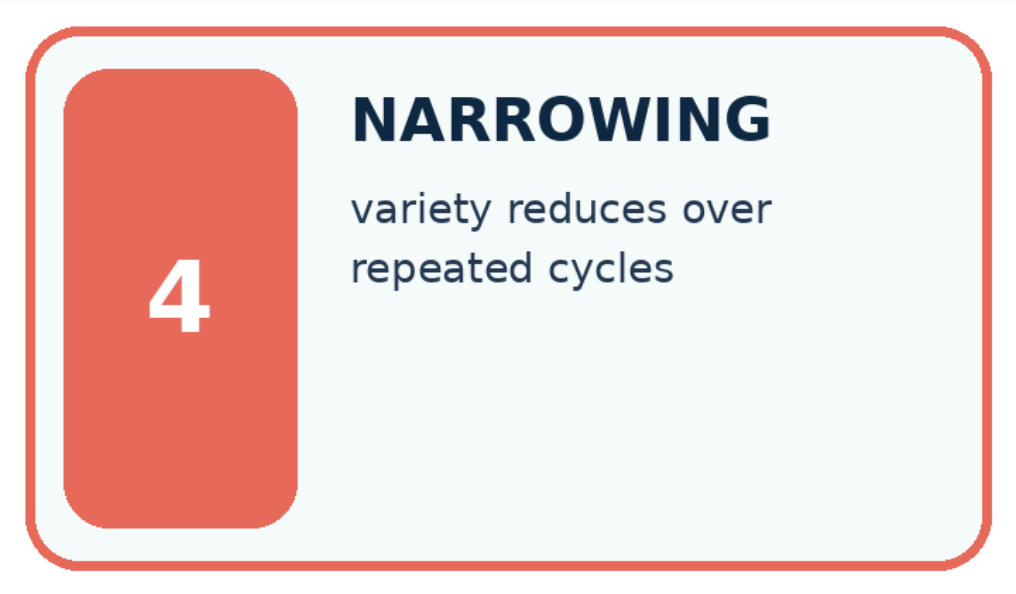
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
FEEDBACK LOOP
output affects later input
2
EXPOSURE
what the user gets a chance to see





Word	My own example	Not this / boundary
feedback loop	A cycle in which outputs affect later inputs and therefore future outputs.	
exposure	The opportunity to see or encounter an item, which can affect later choices.	

Word	My own example	Not this / boundary
interaction data	Records of what people were shown and how they acted, such as views, clicks or corrections.	
narrowing	Reducing the allowed choices, scope or interpretation of a task.	

Words used in the closer explanation

Word	Meaning
Recommender	A procedure that suggests options using a stated policy
Tie	Two options have the same comparison score
Deterministic	The same inputs and rules produce the same result
Explanation	Evidence describing how a result was obtained

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Feedback Loops, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 24 • WORKED EVIDENCE

1. System recommends mostly fantasy.
2. Reader can only click among shown books.
3. Clicks are treated as stronger fantasy preference.
4. Next round shows even more fantasy, hiding other possible interests.

1. System recommends mostly fantasy.

2. Reader can only click among shown books.

3. Clicks are treated as stronger fantasy preference.

4. Next round shows even more fantasy, hiding other possible interests.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Predict the result if the list order is reversed, and if the limit is 8.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Feedback Loops

A fictional catalogue repeatedly recommends two books using observed clicks.

The facts we will use

Supplied facts

Initially each click count is 0. Equal counts choose earlier ID A over B.

Supplied facts

Only the recommended book is shown. In an invented three-round record, A is shown and clicked each round; B is never shown.

AI ACADEMY / AI-2/24

Exposure and clicks after the invented three rounds

	Shown	Clicked
A	3	3
B	0	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/24

Feedback Loops

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial tie

Counts A=0,B=0; ID rule shows A.



2. Record event

A receives a click, so its count becomes 1; B stays 0.



3. Repeat selection

Higher A count causes A to be shown again; after three clicks A=3,B=0.



4. Interpret carefully

B's zero may mean no exposure, not dislike.

CHANGE ONE THING

On a planned trial, show B once and record no click.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Initial tie	Counts A=0,B=0; ID rule shows A.
2. Record event	A receives a click, so its count becomes 1; B stays 0.
3. Repeat selection	Higher A count causes A to be shown again; after three clicks A=3,B=0.
4. Interpret carefully	B's zero may mean no exposure, not dislike.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The selection policy changes which **evidence** is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

What this example does not establish

Keep the stated limit

A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

Keep the stated limit

Read the labelled picture: What extra field would reveal why comparing raw clicks is misleading?

Keep the stated limit

Change one thing: On a planned trial, show B once and record no click. Predict the result before checking the **explanation**. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Feedback Loops claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Clicks reveal complete preference, even for items never shown.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

> **Write a one-sentence explanation of X winning without calling it objectively best.**

STUDENT LESSON · 8 OF 18

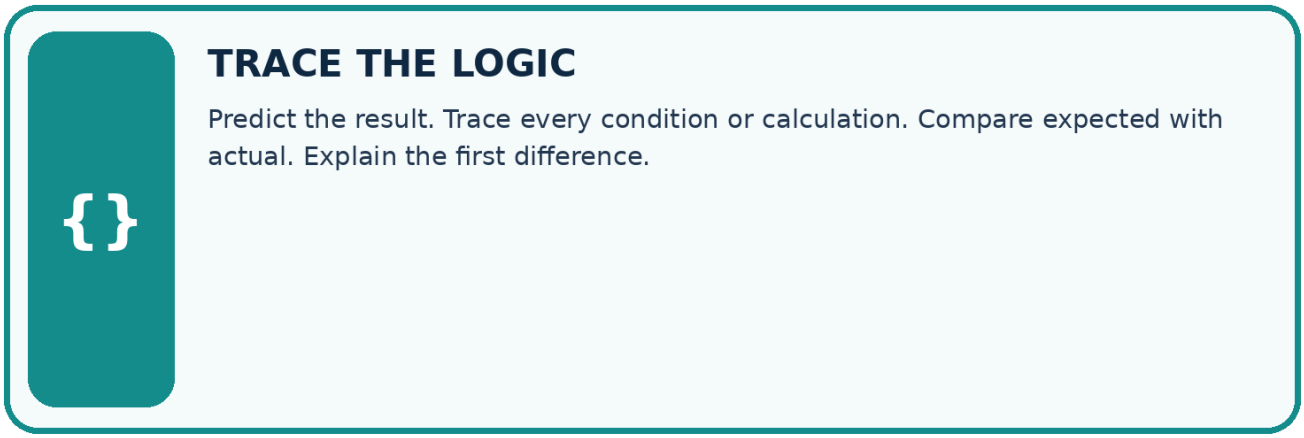
Practise together

> Practise Feedback Loops with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Updating best on score >= best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties. Test both orders and a no-match case after the repair.

Practise with less help: Handle no match

Use workbook W2 for the connected example “Implementing a **recommender**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own **explanation**.

Check your reasoning

Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> What extra field would reveal why comparing raw clicks is misleading?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

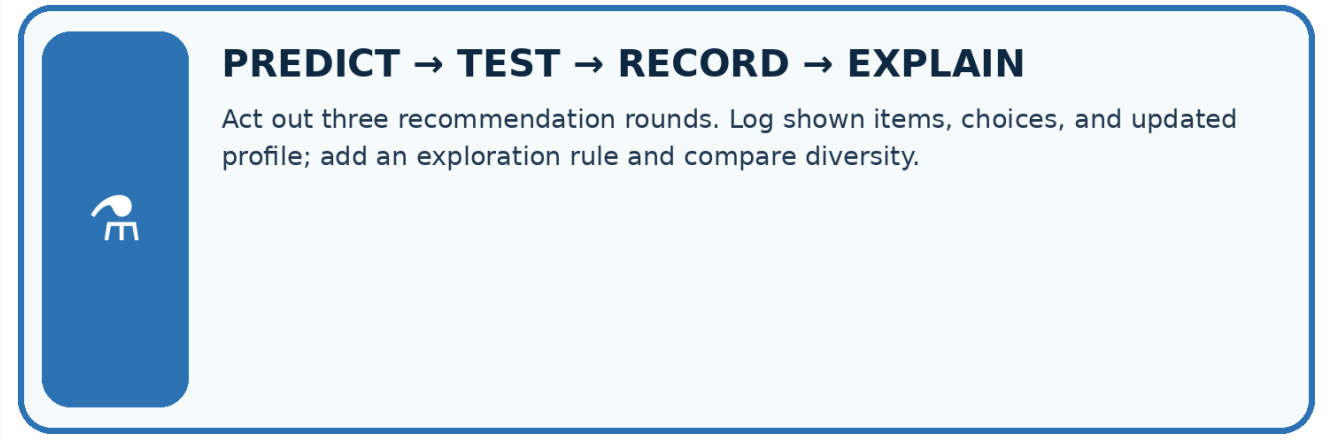
Run the investigation

> Investigate Feedback Loops. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Act out three recommendation rounds. Log shown items, choices, and updated profile; add an exploration rule and compare diversity.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

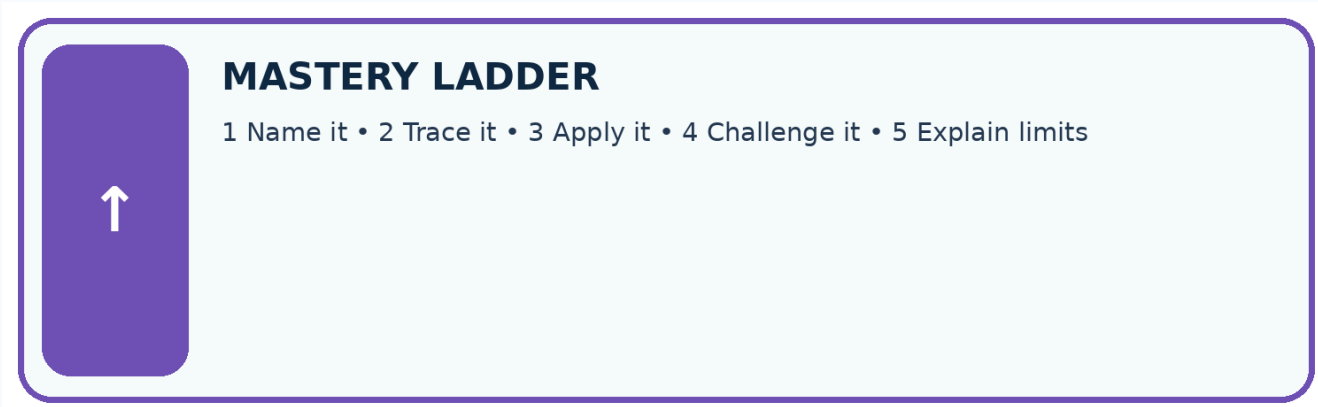
Build your understanding

Move from recall to tracing, diagnosis and transfer in Feedback Loops. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

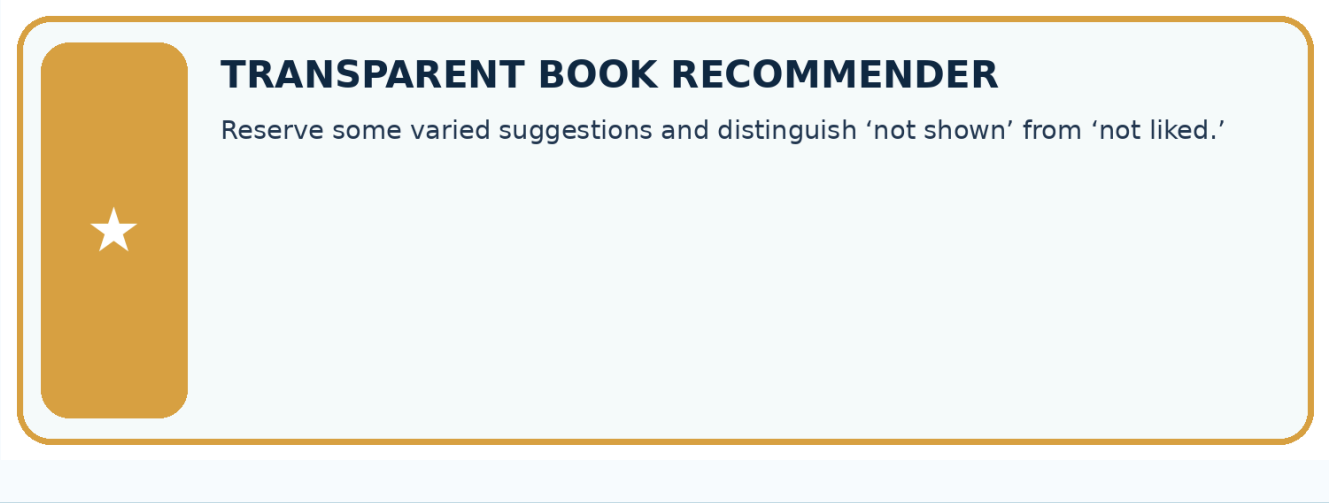
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book **Recommender**

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Reserve some varied suggestions and distinguish 'not shown' from 'not liked.'

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → **explanation** → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**.

Show what you know

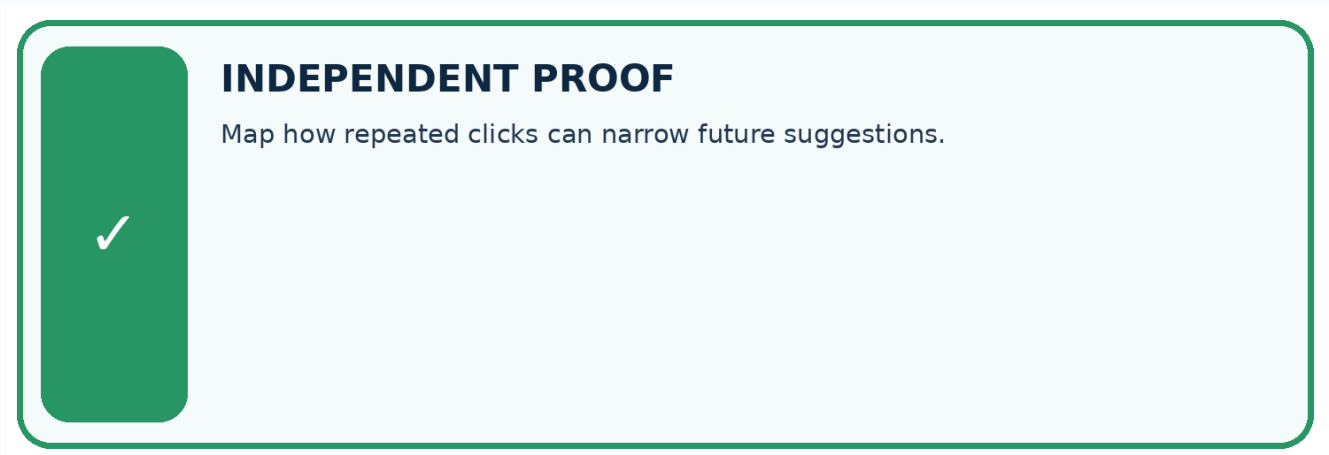
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Map how repeated clicks can narrow future suggestions.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Feedback Loops case for supported practice. Repeated examples do not become fresh assessments.

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A **recommender** shows only adventure books, uses clicks to learn and then shows more adventure.

The question

Explain the **feedback loop**.

Worked explanation

Exposure affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Exposure affects the feedback you collect

Evidence or stage	Value or result
Show adventure	Limits visible choices
Observe clicks	Only among shown items

Evidence or stage	Value or result
Update preferences	Uses those clicks
Show more adventure	Repeats the exposure pattern

A **recommender** shows only adventure books, uses clicks to learn and then shows more adventure.

Explain the **feedback loop**.

Exposure affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 24 Independent application

Close the worked **explanation**. Apply the idea to this different question without copying.

Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Reserve some varied suggestions and distinguish ‘not shown’ from ‘not liked.’

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A **recommender** shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Feedback Loops, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional catalogue repeatedly recommends two books using observed clicks.

By the end, I can	How I will show it
Implement a deterministic recommender	Explain the deciding step, show a trace or test, and state one limit.
Handle ties and no eligible options	Explain the deciding step, show a trace or test, and state one limit.
Explain recommendations with their limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Feedback Loops without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Implement a **deterministic recommender**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue repeatedly recommends two books using observed clicks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B's zero may mean no **exposure**, not dislike.

1. Record event

2. Repeat selection

3. Initial tie

4. Interpret carefully

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported **explanation**: Interpret carefully B's zero may mean no **exposure**, not dislike. The selection policy changes which **evidence** is available.

A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: On a planned trial, show B once and record no click. Which conclusion is supported by the supplied facts?

1. B gains one **exposure** but still zero clicks.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported **explanation**: B gains one **exposure** but still zero clicks. The **observation** now supports a nonclick on that display, not a permanent dislike.

Repair a missing building block

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing..

Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

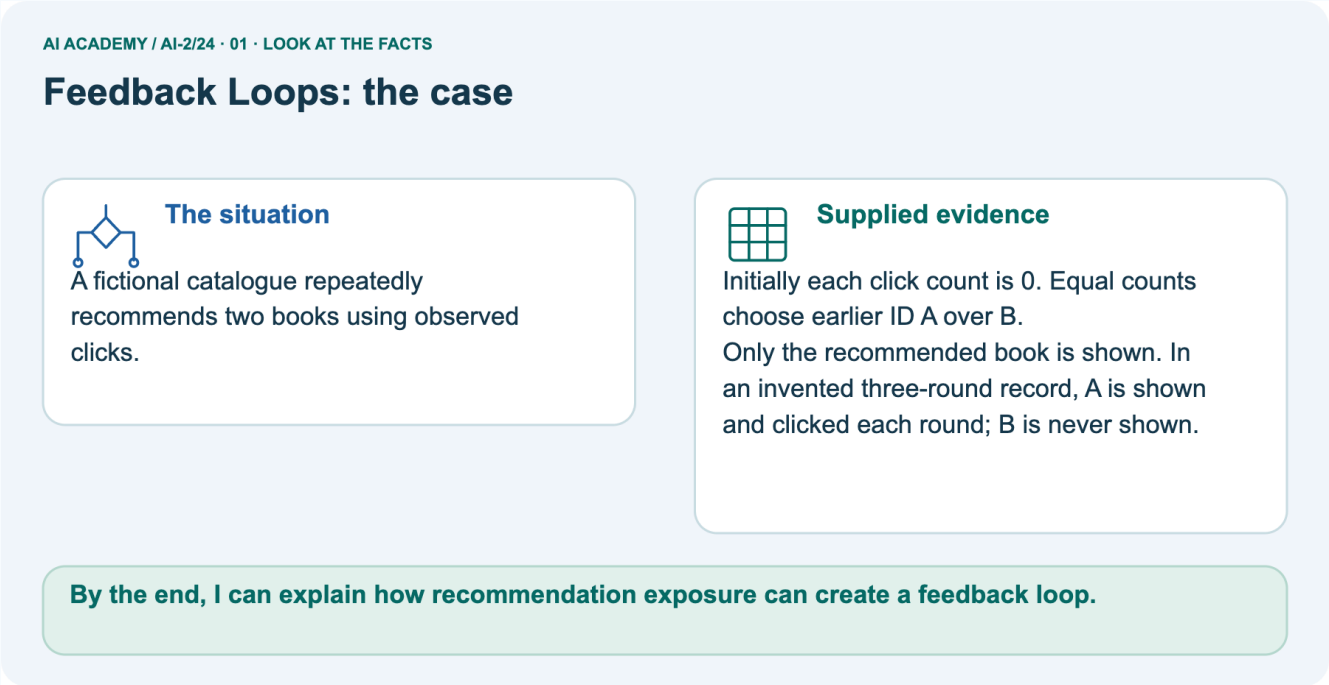
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Feedback Loops graphic using its labels, deciding evidence and relevant limitation.

01. Feedback Loops: the case



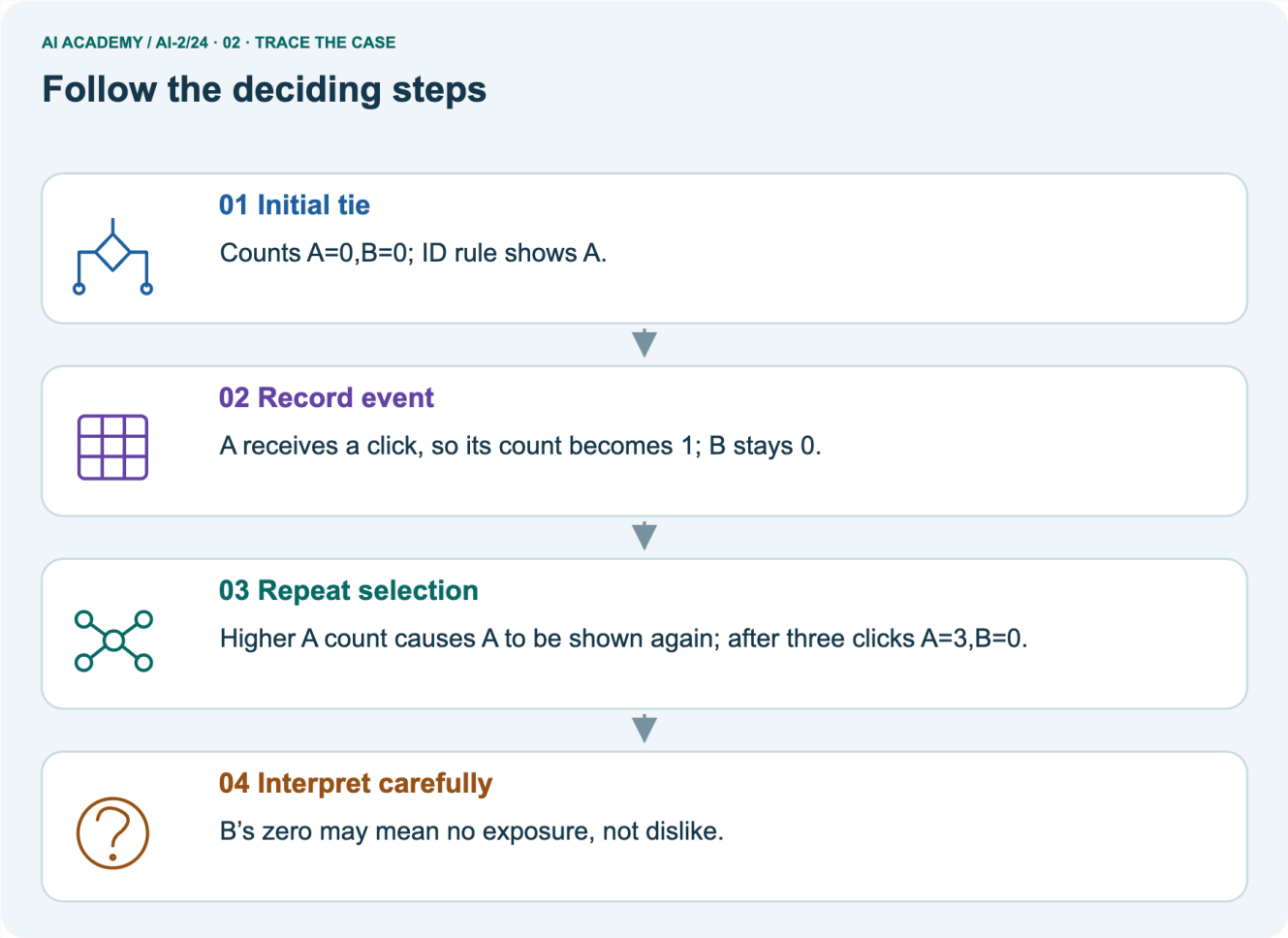
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue repeatedly recommends two books using observed clicks. Supplied facts: Initially each click count is 0. Equal counts choose earlier ID A over B. Only the recommended book is shown.

In an invented three-round record, A is shown and clicked each round; B is never shown.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

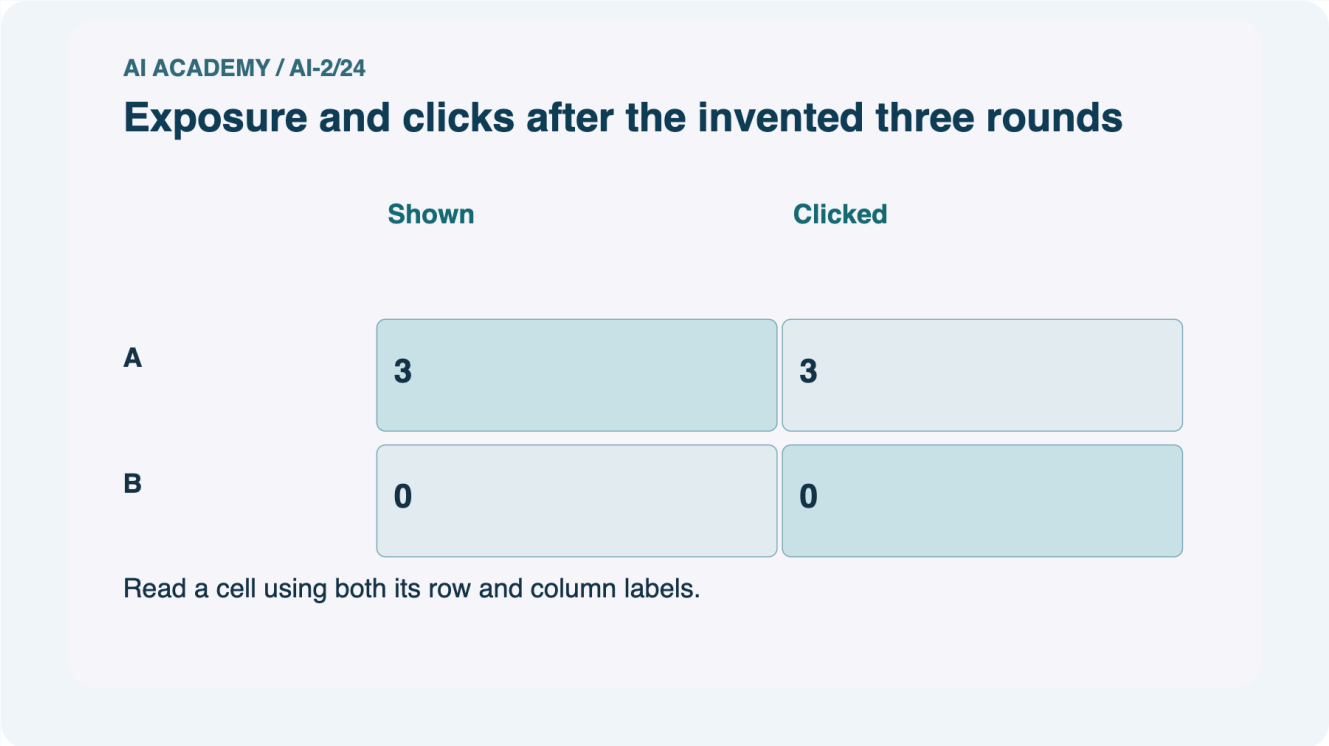


Follow the labelled picture with the matching explanation. Open the vector diagram

Initial tie: Counts A=0,B=0; ID rule shows A. → Record event: A receives a click, so its count becomes 1; B stays 0. → Repeat selection: Higher A count causes A to be shown again; after three clicks A=3,B=0. → Interpret carefully: B's zero may mean no **exposure**, not dislike.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Exposure and clicks after the invented three rounds

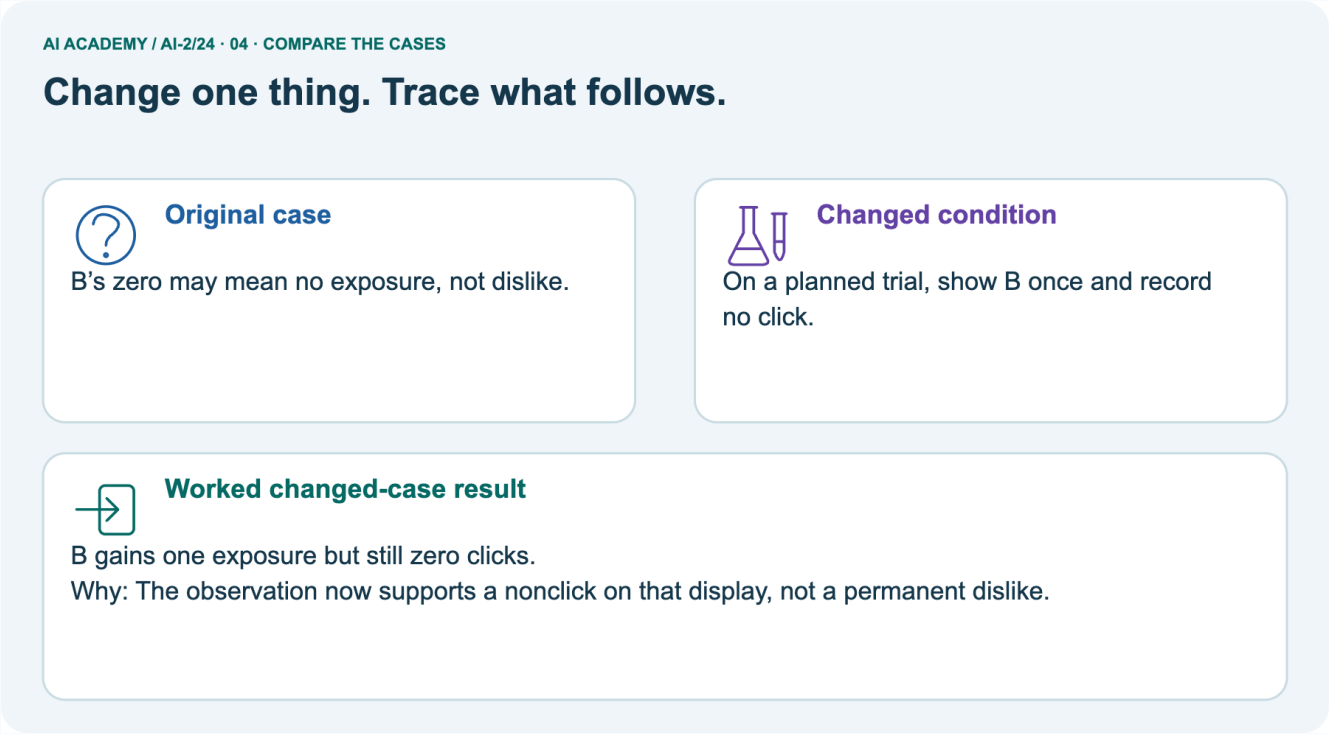


Follow the labelled picture with the matching explanation. Open the vector diagram

Exposure and clicks after the invented three rounds. A: Shown = 3, Clicked = 3. B: Shown = 0, Clicked = 0.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



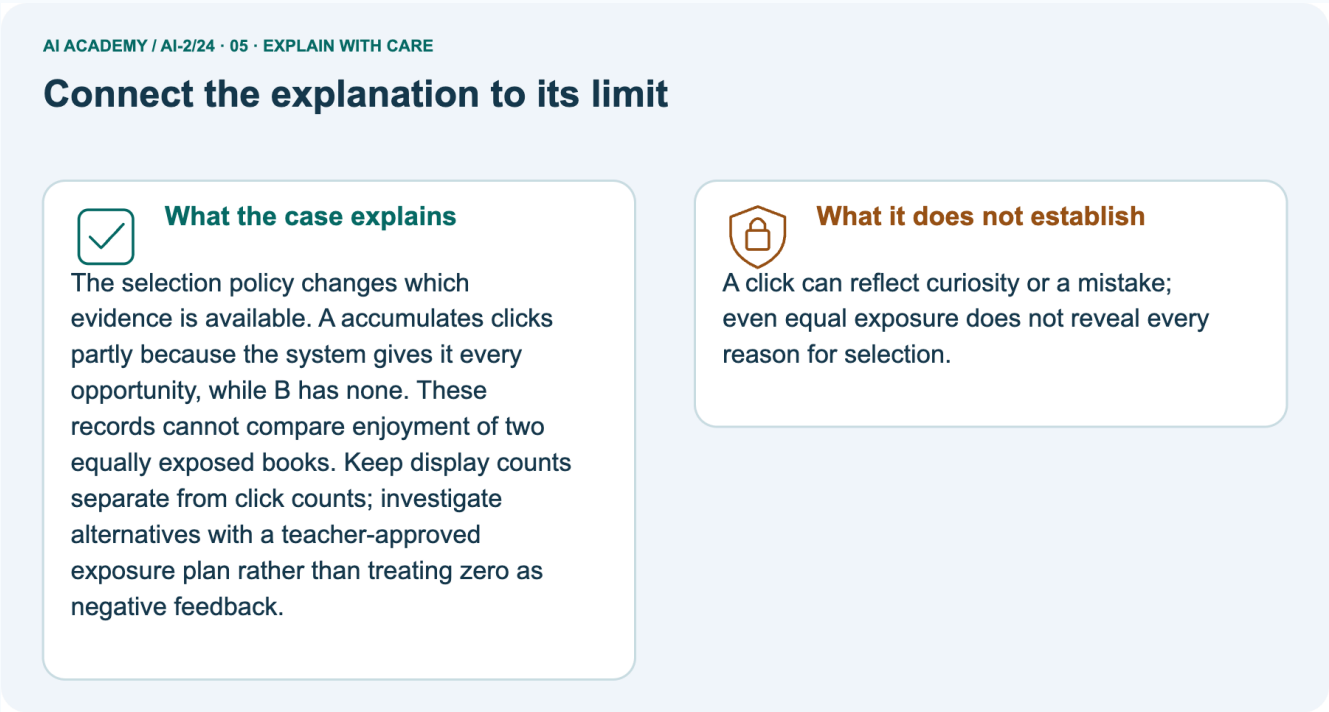
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B's zero may mean no **exposure**, not dislike. Changed condition: On a planned trial, show B once and record no click. Expected result: B gains one **exposure** but still zero clicks.

Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

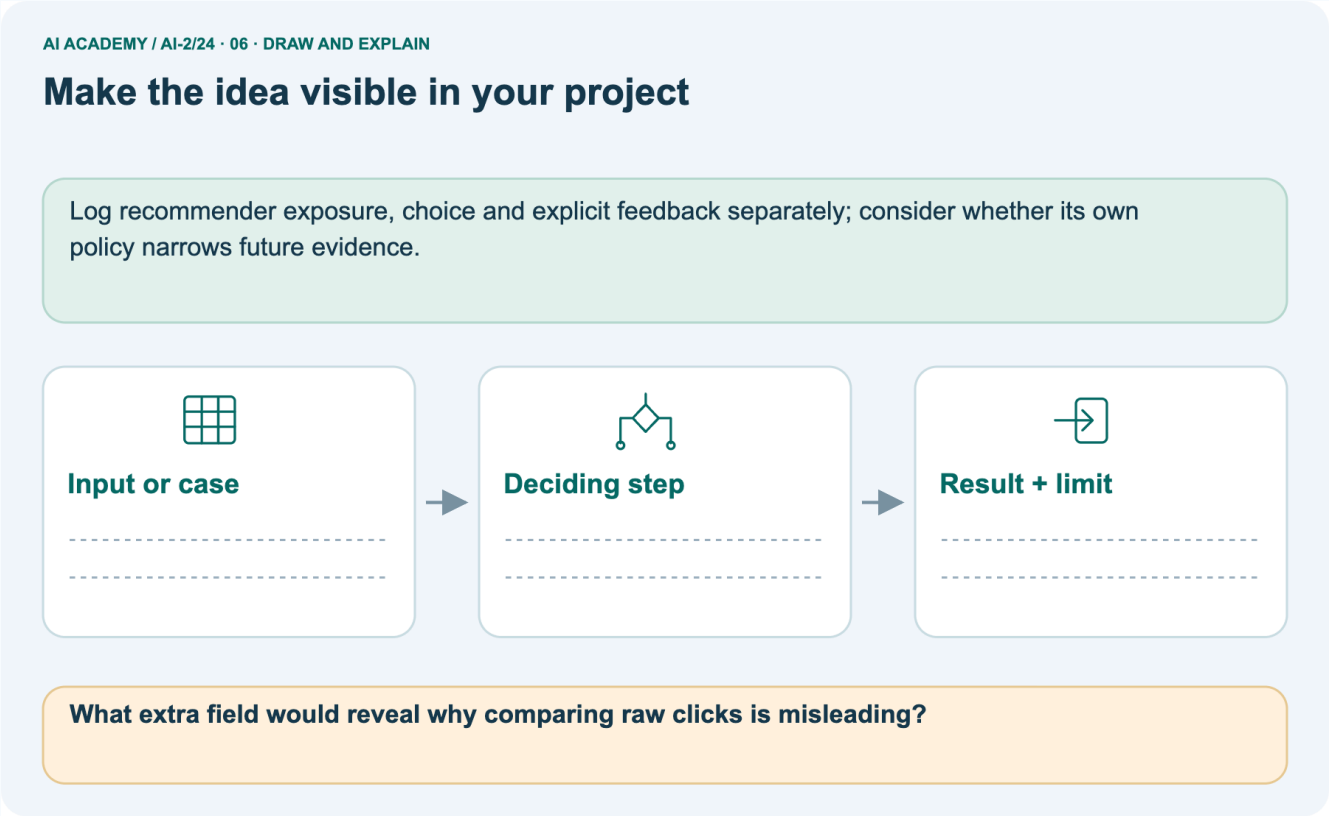
The selection policy changes which **evidence** is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

Limit: A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: What extra field would reveal why comparing raw clicks is misleading?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.